

One of the problems I face in life is that whenever I tell people that the Gaussian bell curve is not ubiquitous in real life, only in the minds of statisticians, they require me to “prove it”—which is easy to do, as we will see in the next two chapters, yet nobody has managed to prove the opposite. Whenever I suggest a process that is not Gaussian, I am asked to justify my suggestion and to, beyond the phenomena, “give them the theory behind it.” We saw in [Chapter 14](#) the rich-get-richer models that were proposed in order to justify not using a Gaussian. Modelers were forced to spend their time writing theories on possible models that generate the scalable—as if they needed to be apologetic about it. Theory shmeory! I have an epistemological problem with that, with the need to justify the world’s failure to resemble an idealized model that someone blind to reality has managed to promote.

My technique, instead of studying the possible models generating non-bell curve randomness, hence making the same errors of blind theorizing, is to do the opposite: to know the bell curve as intimately as I can and identify where it can and cannot hold. I know where Mediocristan is. To me it is frequently (nay, almost always) the users of the bell curve who do not understand it well, and have to justify it, and not the opposite.

This ubiquity of the Gaussian is not a property of the world, but a problem in our minds, stemming from the way we look at it.

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The next chapter will address the scale invariance of nature and address the properties of the fractal. The chapter after that will probe the misuse of the Gaussian in socioeconomic life and “the need to produce theories.”

I sometimes get a little emotional because I’ve spent a large part of my life thinking about this problem. Since I started thinking about it, and conducting a variety of thought experiments as I have above, I have not for the life of me been able to find anyone around me in the business and statistical world who was intellectually consistent in that he both accepted the Black Swan and rejected the Gaussian and Gaussian tools. Many people accepted my Black Swan idea but could not take it to its logical conclusion, which is that you cannot use one single measure for randomness called standard deviation (and call it “risk”); you cannot expect a *simple* answer to characterize uncertainty. To go the extra step requires courage, commitment, an ability to connect the dots, a desire to understand randomness fully. It also means not accepting other people’s wisdom as gospel. Then I started finding physicists who had rejected the Gaussian tools but fell for another sin: gullibility about precise predictive models, mostly elaborations around the preferential attachment of